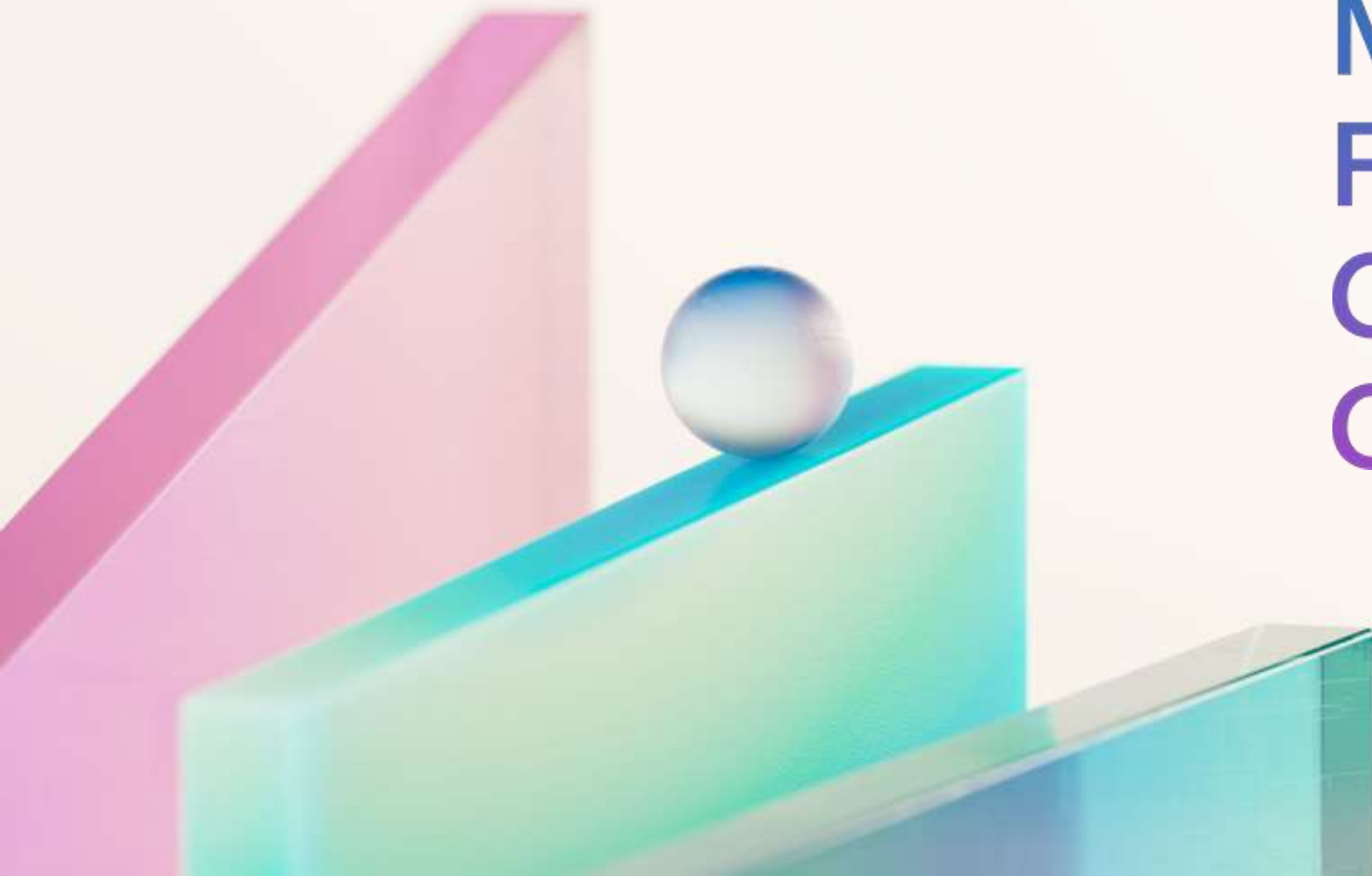
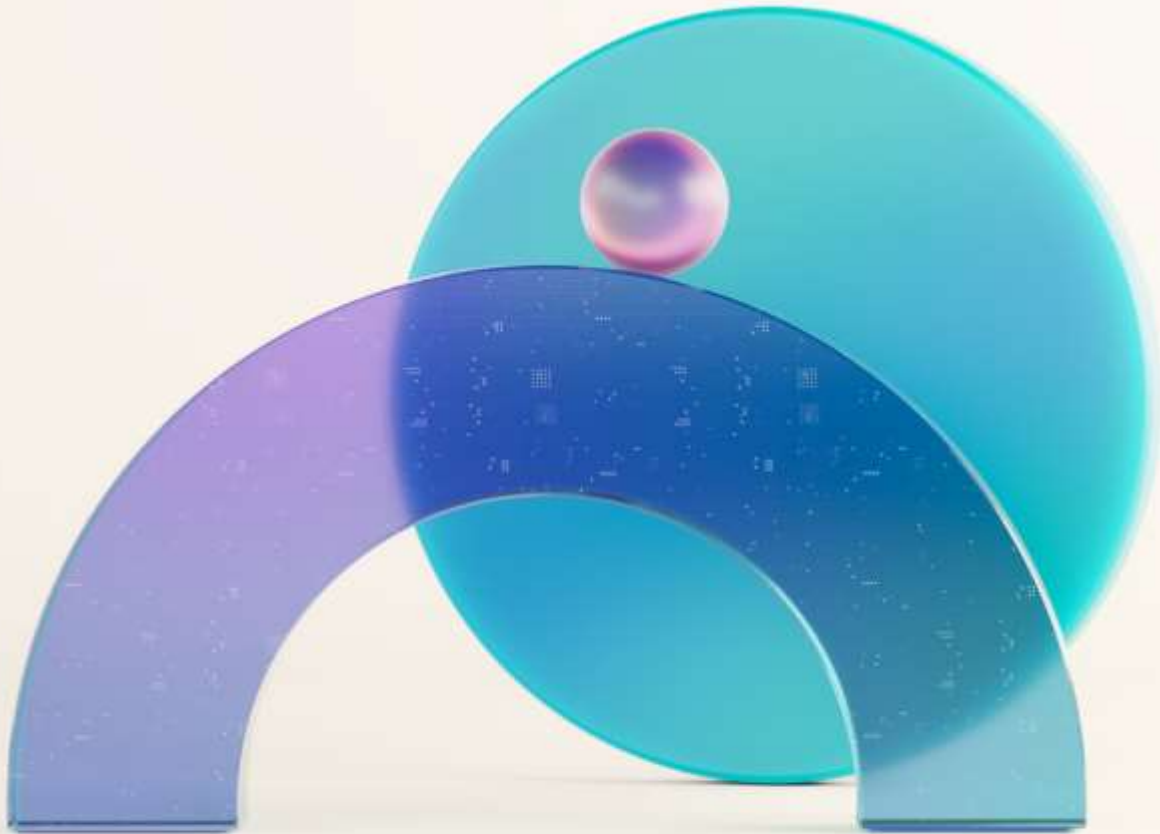




Microsoft Fabric Community Conference





A Live Demonstration of Microsoft Fabric Lakehouse

Transforming Data into Gold

Agenda

Fabric is Fantastic -
but also
overwhelming



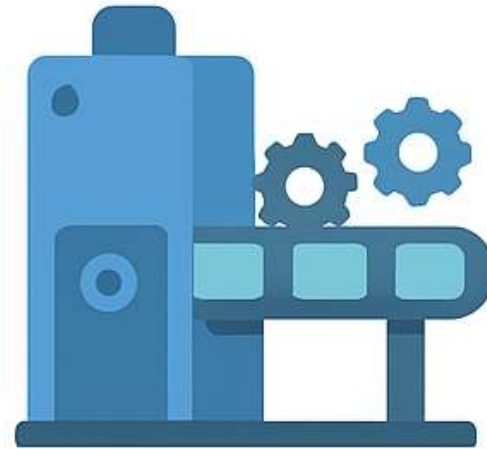
Lakehouse
starting-point
architecture



Kickstart
the
Lakehouse
Setup



Start the
Lakehouse
with an
accelerator



Extend and
customize the
accelerator



Just Blindbæk

 /blindbaek

Peer Grønnerup

 /groennerup

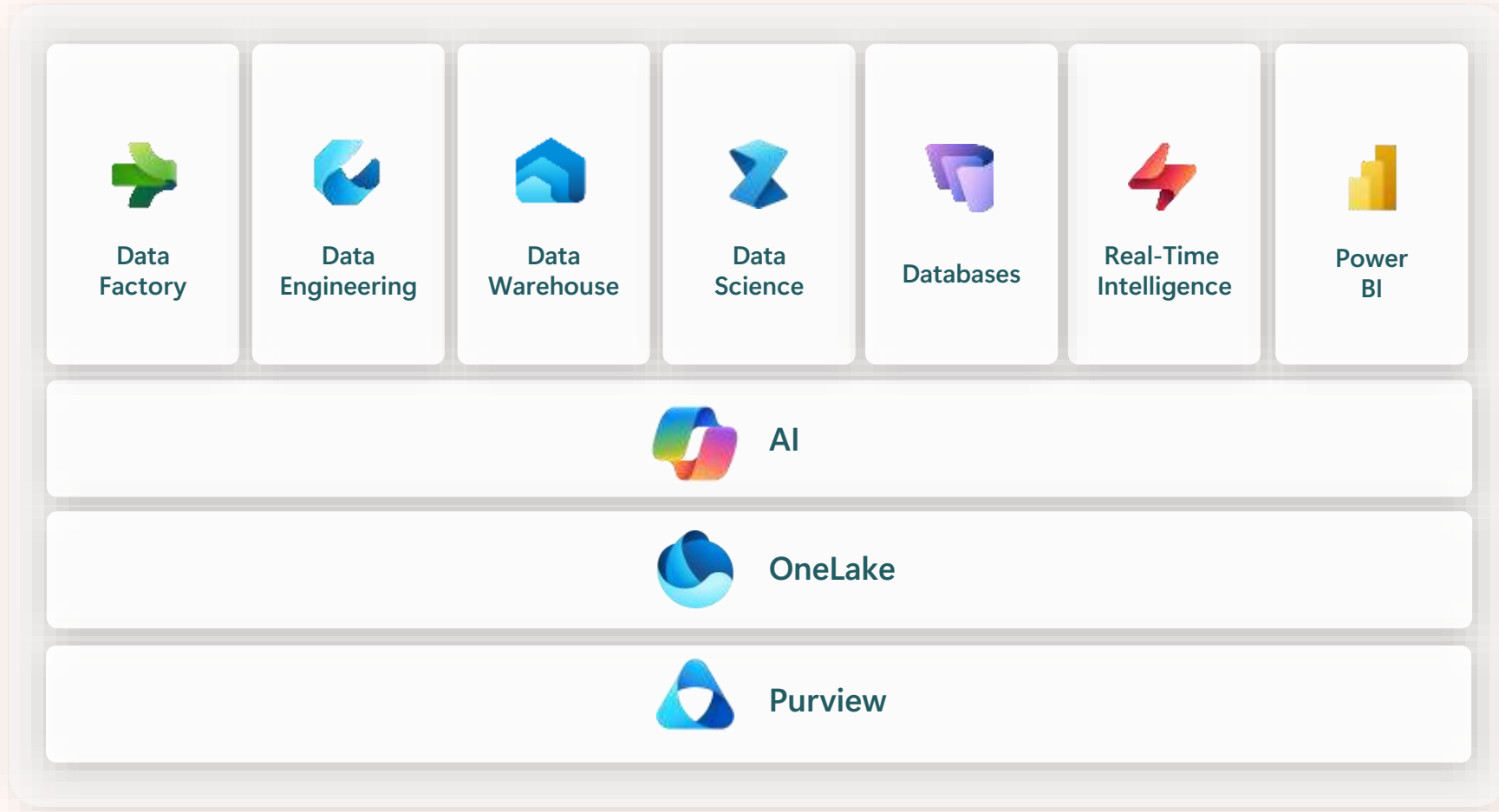
Henrik Reich

 /henrik-reich



Principal Architects at twoday Data & AI in Denmark

Fabric: End-to-end unified analytical platform



Software as a Service

Seven workloads

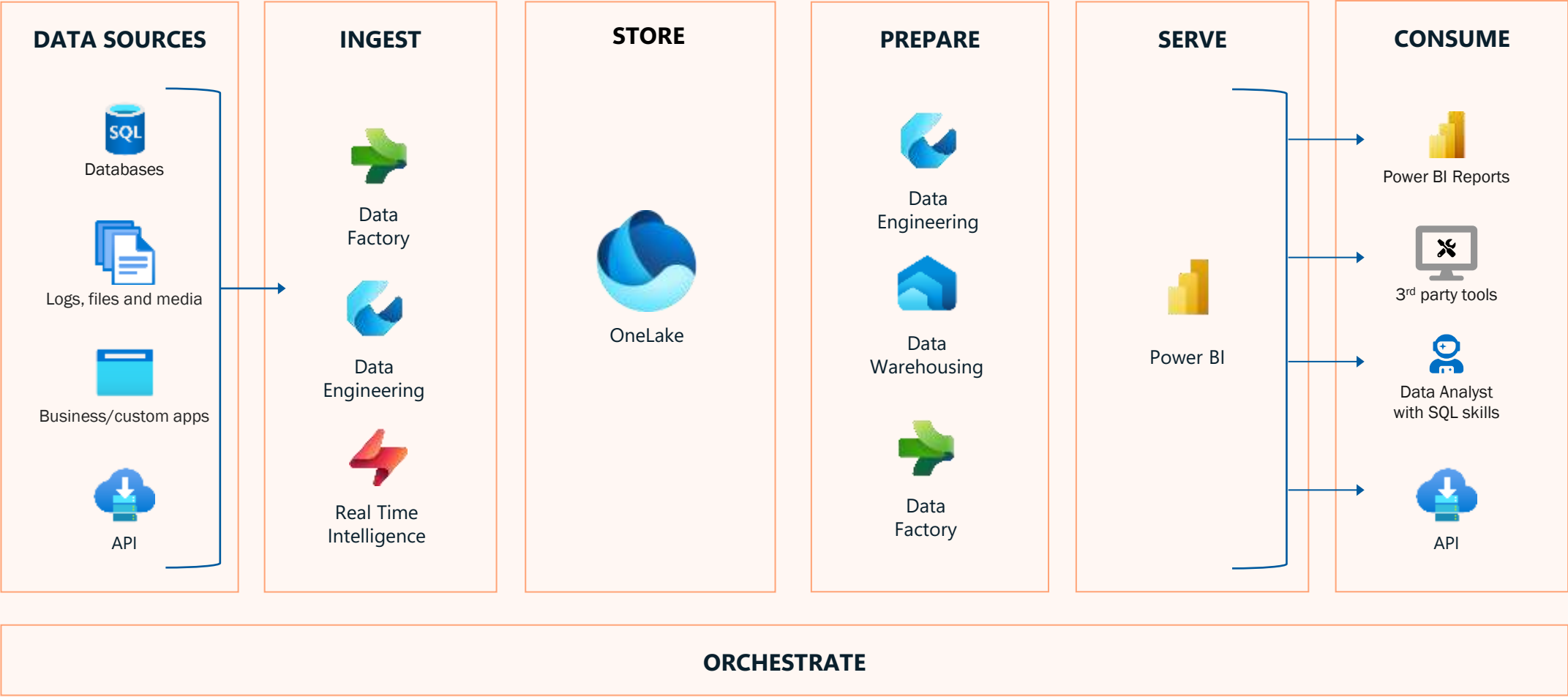
- Low-code/no-code
- Pro developer / engineers

- **30+ items**

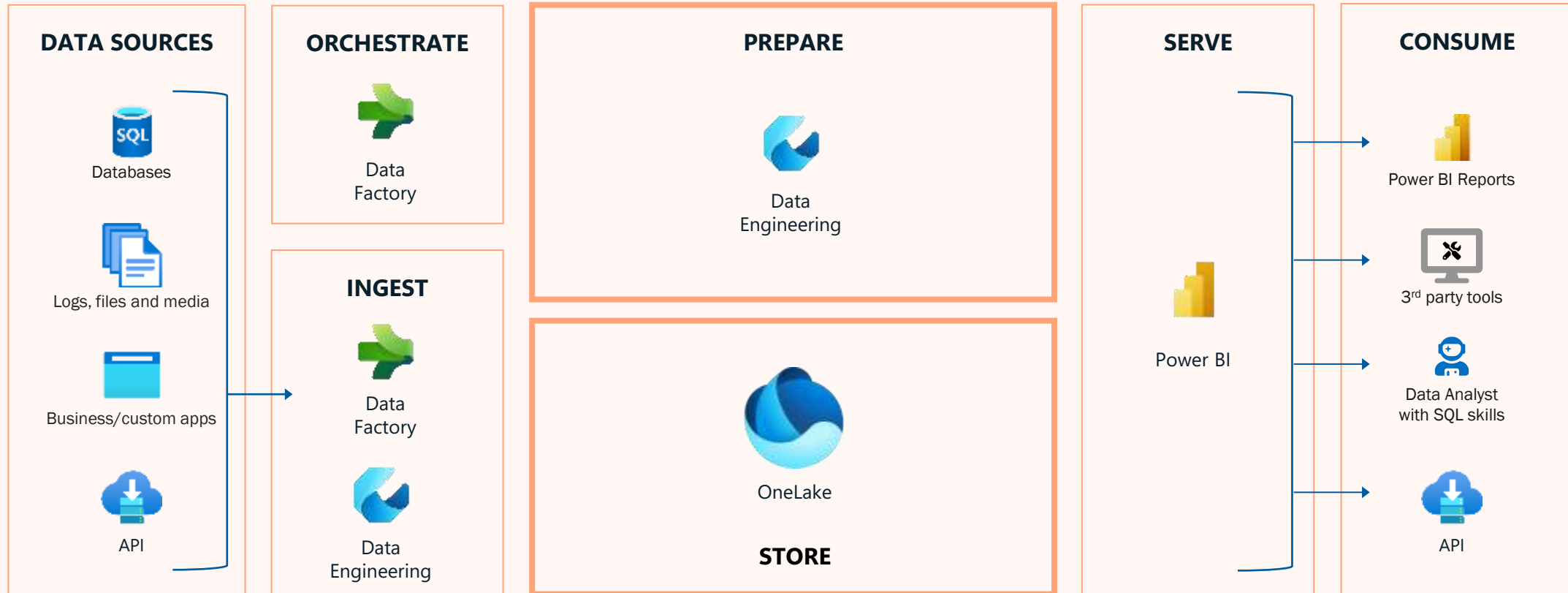
OneLake & Delta format

- Separation of storage and compute

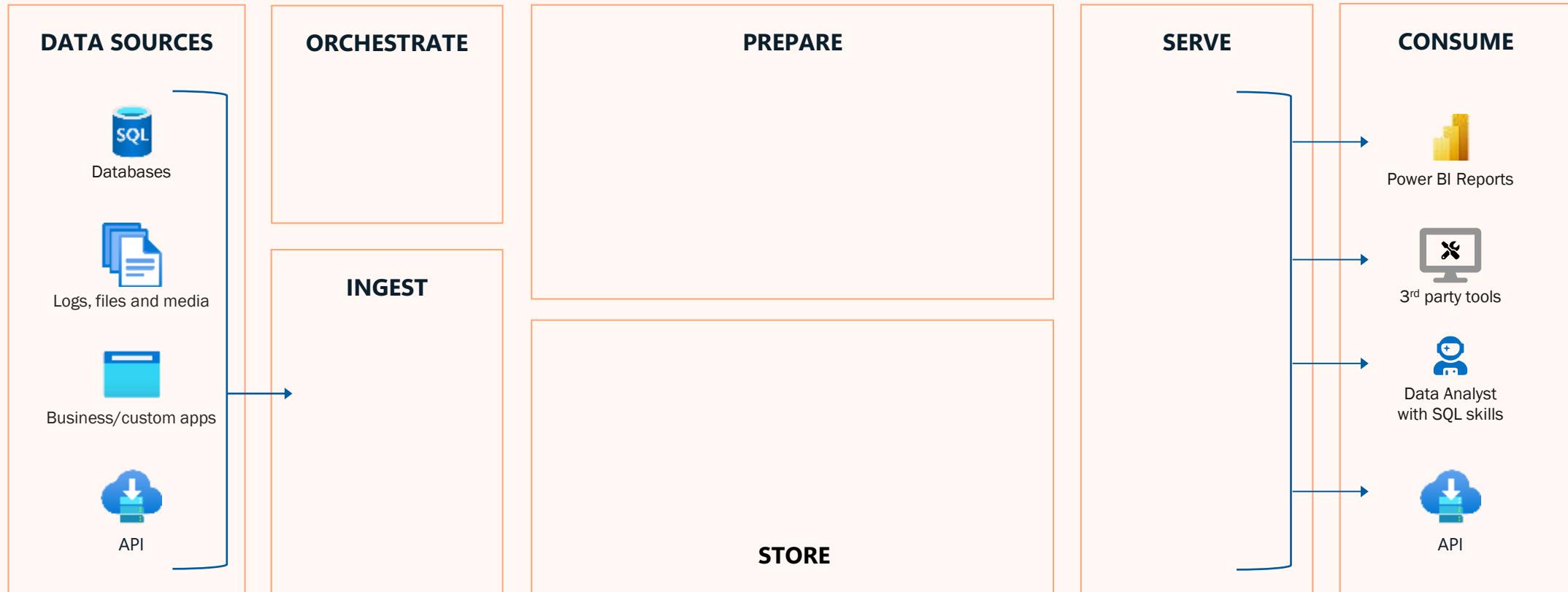
Architecture view of Fabric workloads



Lakehouse view of Fabric workloads



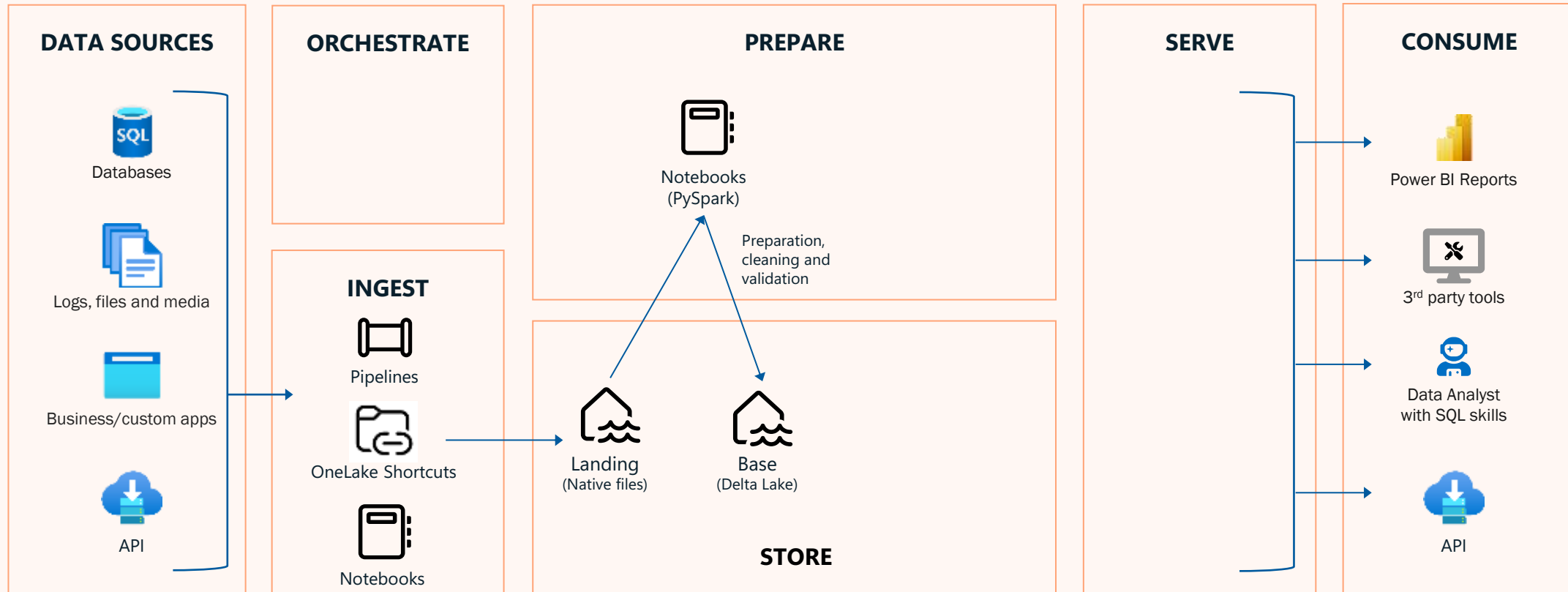
Dataflow in a Data Lakehouse in Fabric



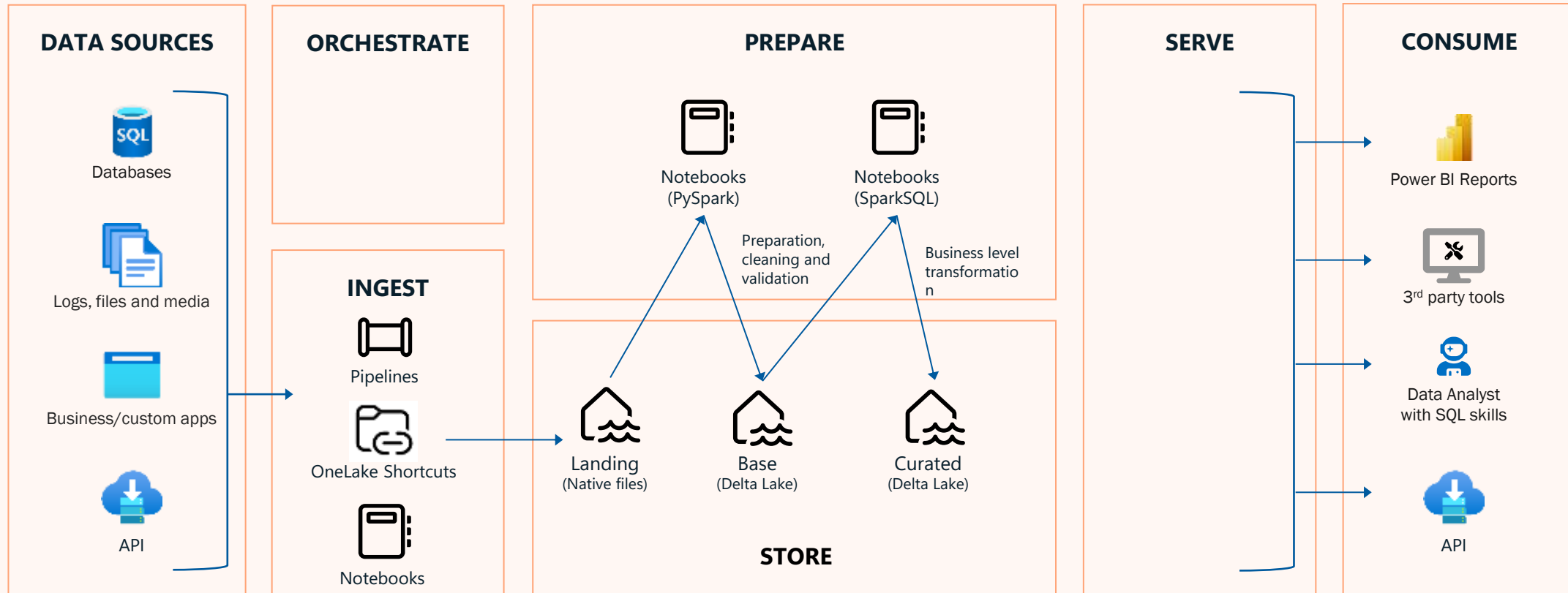
Dataflow in a Data Lakehouse in Fabric



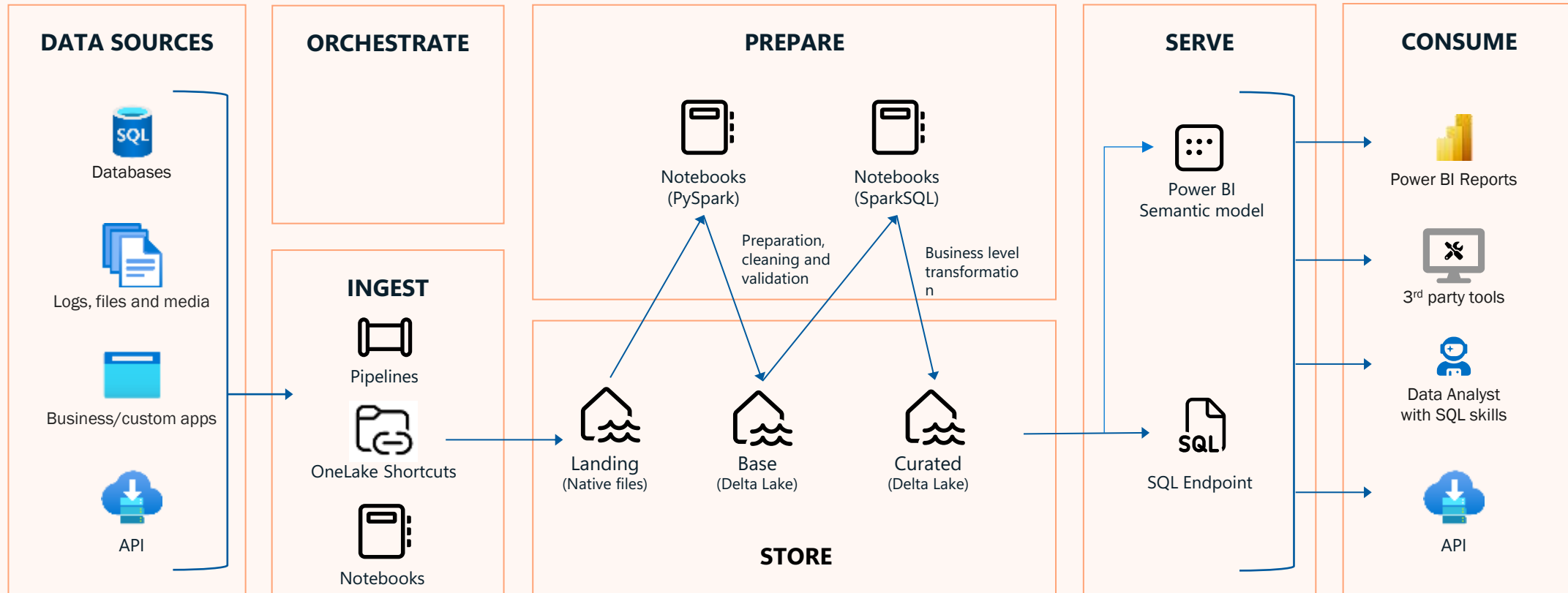
Dataflow in a Data Lakehouse in Fabric



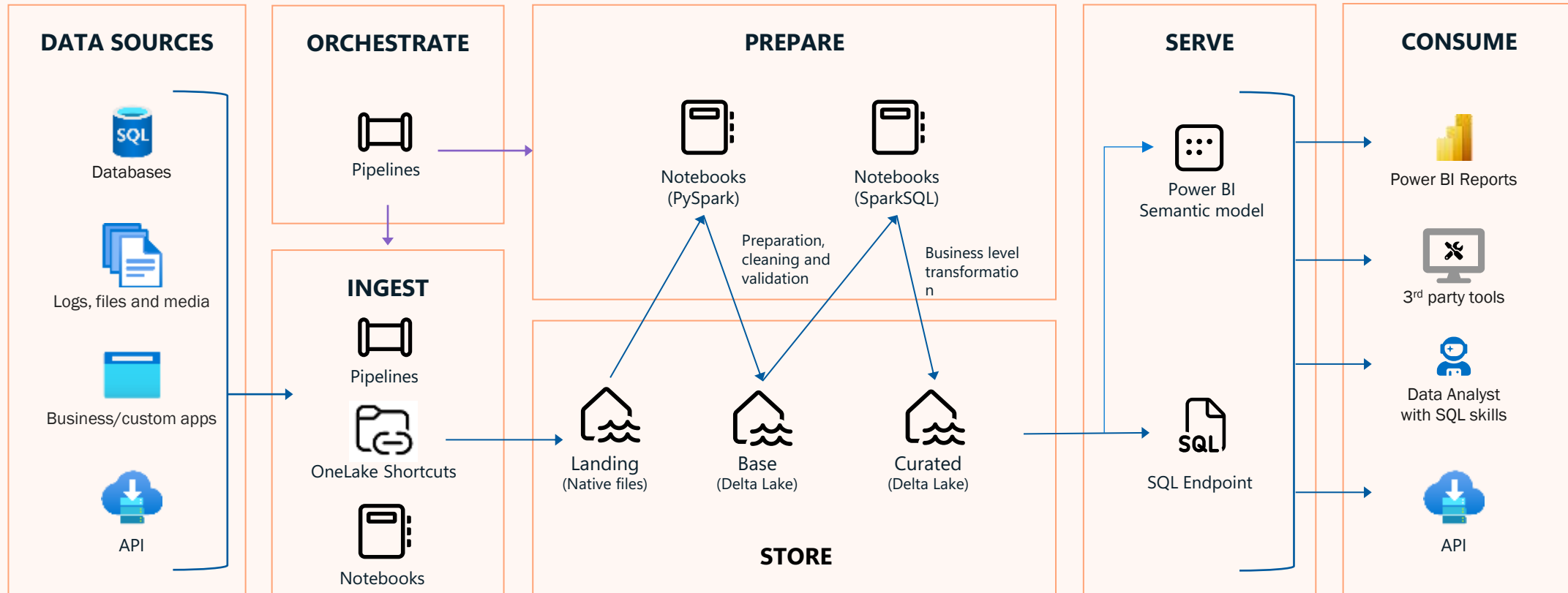
Dataflow in a Data Lakehouse in Fabric



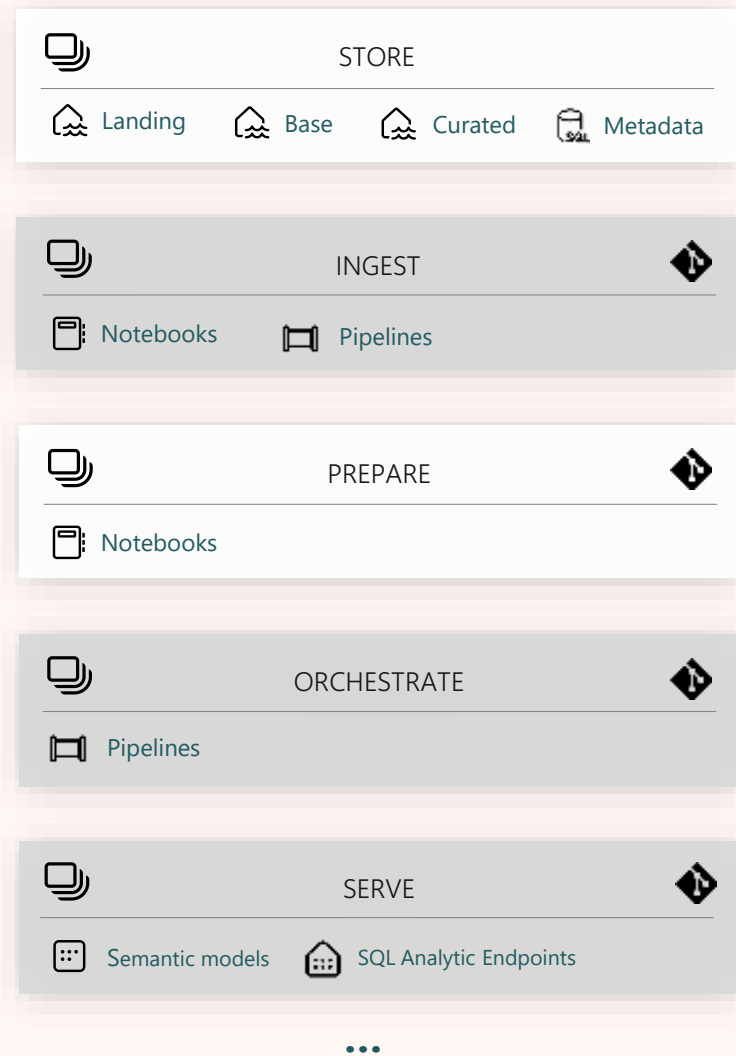
Dataflow in a Data Lakehouse in Fabric



Dataflow in a Data Lakehouse in Fabric

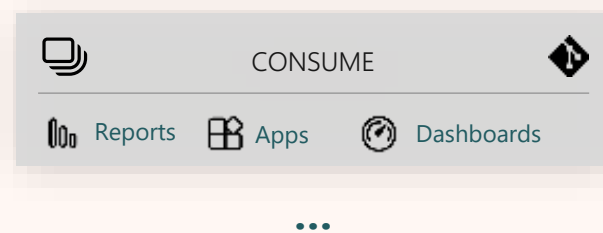


Fabric Workspace structure



Layer separated workspace pattern

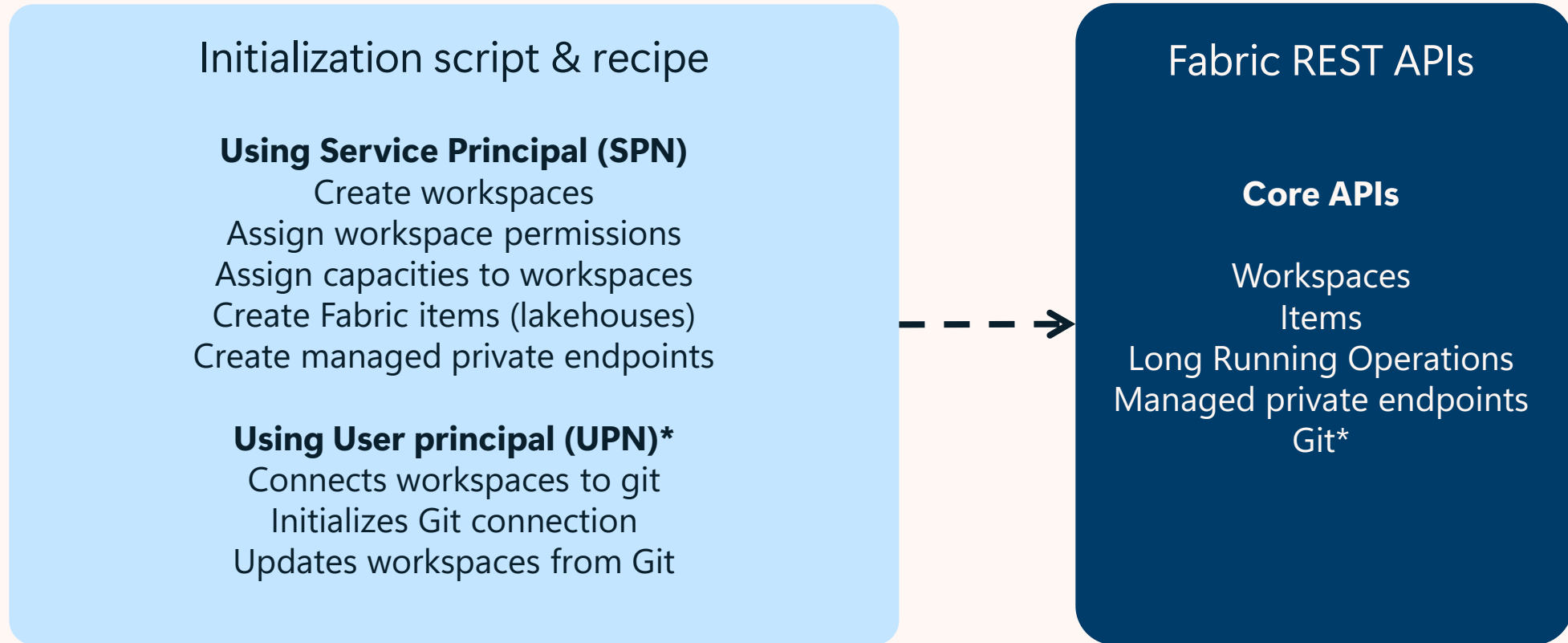
- Item organization
- Enhanced access control
- Support capacity separation



Git integration

- Single repository
- Layer specific folders
- Additional folders for automation, documentation etc.

Setup the Fabric Lakehouse structure in 1 minute!



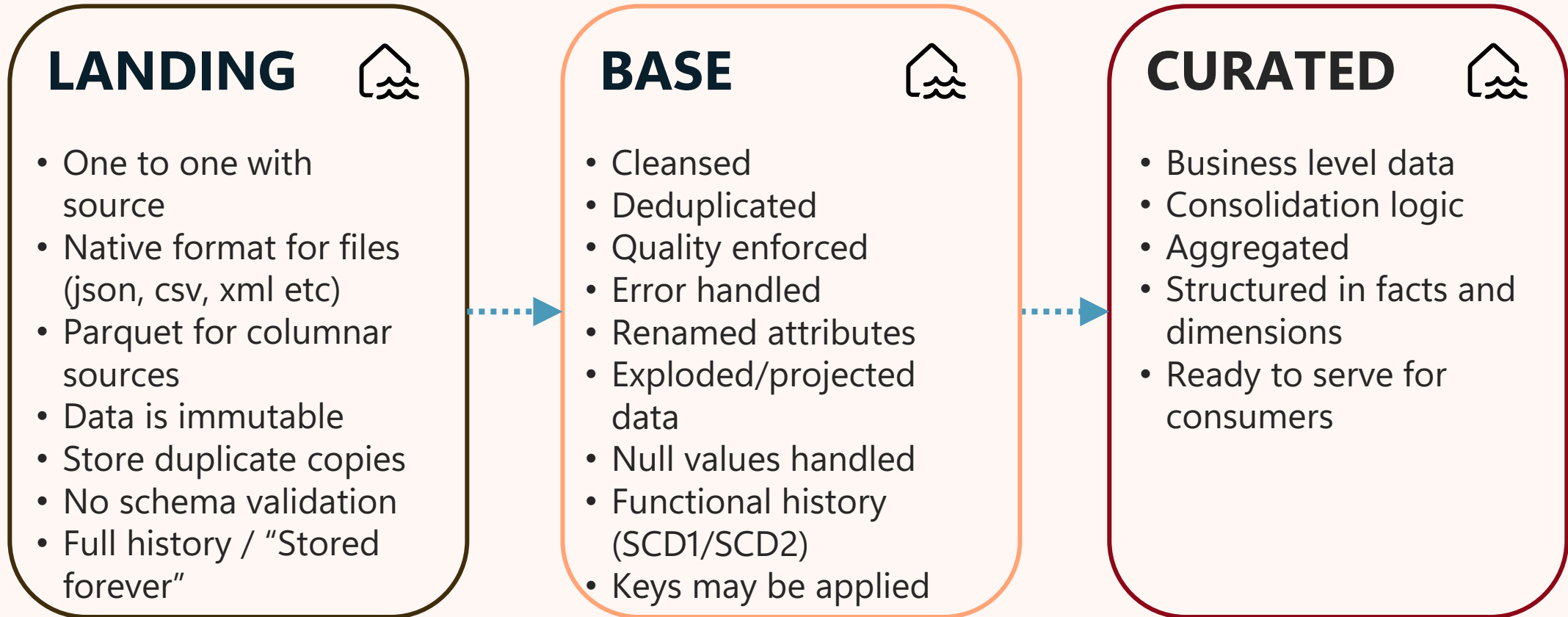
Visit Peer's blog post "**Automating Fabric: Kickstart your Fabric Setup with Python and Fabric REST APIs**" on <https://peerinsights.hashnode.dev>

AquaShack: Data Lakehouse accelerator



- Pico-example of a meta-data driven lakehouse for Microsoft:
 - Speeds up setup
 - Teaches best practices
 - Clean architecture
- Classic Data Lakehouse with three layers, where data is moved between.
 - Landing: Holds the data in native format. Relation sources are stored in parquet.
 - Base: Aligned clean data, all stored in Delta tables.
 - Curated: Data for serving, business logic is applied, star schemas are defined etc.
- The notebooks builds upon functions defined in AquaShack_functions.
- We are reading metadata from JSON, it could have been from a database or YAML file as well. The important part is it ends in a common structure.

Layers in the Data Lakehouse



Metadata driven from Landing to Base

- The **AquaShack Functions** notebook contains general functions used for operations like reading DataFrames, writing DataFrames, and utilities for manipulating DataFrames. To use the functions, include the notebook in your Spark session with the %run command. Once included, you can reference any function within the notebook, such as read_meta_from_json, write_to_delta_overwrite, and others.
- Data movement between the Landing and Base layers is managed by the LandingToBase notebook. This notebook reads metadata from a designated metadata json-file and processes the source path (folder) specified within it.
- Metadata is maintained in the meta_data.json file. This metadata defines the rules for reading data from the Landing layer and writing it to the Base layer.

```
[
  {
    "source": "Files/External/Sample data/Sales/Customers/Customers.csv",
    "format": "csv",
    "destination": "sales_Customers",
    "projected_columns": ["CustomerID", "FirstName", "LastName"]
  },
  {
    "source": "Files/External/Sample data/Sales/Products/Products.parquet",
    "format": "parquet",
    "destination": "sales_Products",
    "projected_columns": []
  },
  {
    "source": "Files/External/Sample data/Sales/Transactions/",
    "format": "csv",
    "destination": "sales_Transactions",
    "projected_columns": []
  }
]
```

Base to Curated

- The Curated layer stores data in a dimensional model with entities like dimensions, facts, and bridge tables.
- Entity-specific Spark notebooks handle data creation, named as:
 - Load_Dimension_[dimension_name]
 - Load_Fact_[fact_name]
- Each notebook follows a standard structure:
 - Parameters & Settings.
 - Business Logic: Written in SparkSQL.
 - Generic Load Execution: Functions like load_dimension or load_fact.
- All notebooks are tied to the Base Lakehouse item for centralized management.

Defining settings

Some settings run through the notebook and easy to set once before the job starts.

```
1 destination_lakehouse: str = 'Curated'
2
3 dimension_name: str = 'Product'
```

Defining business requirements

This could also be pure PySpark, it depends on the skill set of the BI Developer.

```
1 --Sql
2 CREATE OR REPLACE TEMPORARY VIEW dim_df AS
3
4 SELECT
5     -- Keys
6     Id as DimD_Product,
7
8     -- BusinessKey
9     ProductId as ProductsId,
10
11     -- Attributes
12     Manufacturer,
13     ProductName,
14     Price
15
16 FROM
17     Base.sales_products
```

Writing the dimension

Fetch the DataFrame and pass it to our function for writing dimensions

```
1 dim_df = spark.table('dim_df')
2
3 load_dimension(df = dim_df, database = destination_lakehouse, table = dimension_name)
```

Overwriting Curated.dim_sales_products

Quick start guide to AquaShack



- **Go to GitHub:** <https://github.com/ChristianHenrikReich/AquaShack>
- Download or copy the Setup.ipynb notebook that holds all to get started and set up the AquaShack lakehouse example.
- The only required action is to import and run the notebook within the target Fabric Workspace.
- The setup notebook will create all you need:
 - The three Lakehouses
 - All the Notebooks – attached to default lakehouses
 - Sample metadata file
 - Sample data in Landing

Extend and Customize the Accelerator

- AquaShack is a pico example - a small but powerful starting point.
- Teach the medallion architecture, lakehouse concepts, and hands-on Spark development in Fabric
- Get you going fast and help you build good habits from day one

Extend with more features

- Automated setup: Full setup of infrastructure, workspaces and items
- Metadata: Integration, processing and orchestration with metadata
- Ingestion: Batch ingest from multiple sources using delta and full.
- Common transformations: Deduplication, flattening of complex data structures, data conversion, translations and more.
- Update strategies: Append, overwrite, upsert type 1 & 2 etc.
- Data warehousing: Functions specifically for dimensional modelling
- Samples: Sample implementations for all functionality and layers
- Orchestration: Through Pipelines and Notebooks
- Deployment: Full enterprise CI/CD using Azure DevOps Pipelines





Extend with more modules

Infrastructure

Publisher app and scripts

- Azure infrastructure
- Fabric env. setup



Solution

Metadata-driven ingest, transform and orchestration

- Data Pipelines
- Notebooks
- Onelake as storage
- Metadata-driven development



CI/CD

Pipelines and scripts

- Build and release
- Handling features



Documentation, learning and guidelines

Recommended sessions

Tuesday

 **1:45 - 2:45 PM** |  **Grand BR 114**

From Setup to CI/CD: Automating Microsoft Fabric for Scalable Data Solutions

Peer Grønnerup

Wednesday

 **11:15 AM - 12:15 PM** |  **Grand BR 122**

Automating CI/CD Processes in Microsoft Fabric: Best Practices and Latest Developments

Jacob Knightly & Nimrod Shalit

 **2:00 - 3:00 PM** |  **Grand BR 120**

Building a Metadata-driven Framework with Fabric SQL Database and Data Factory

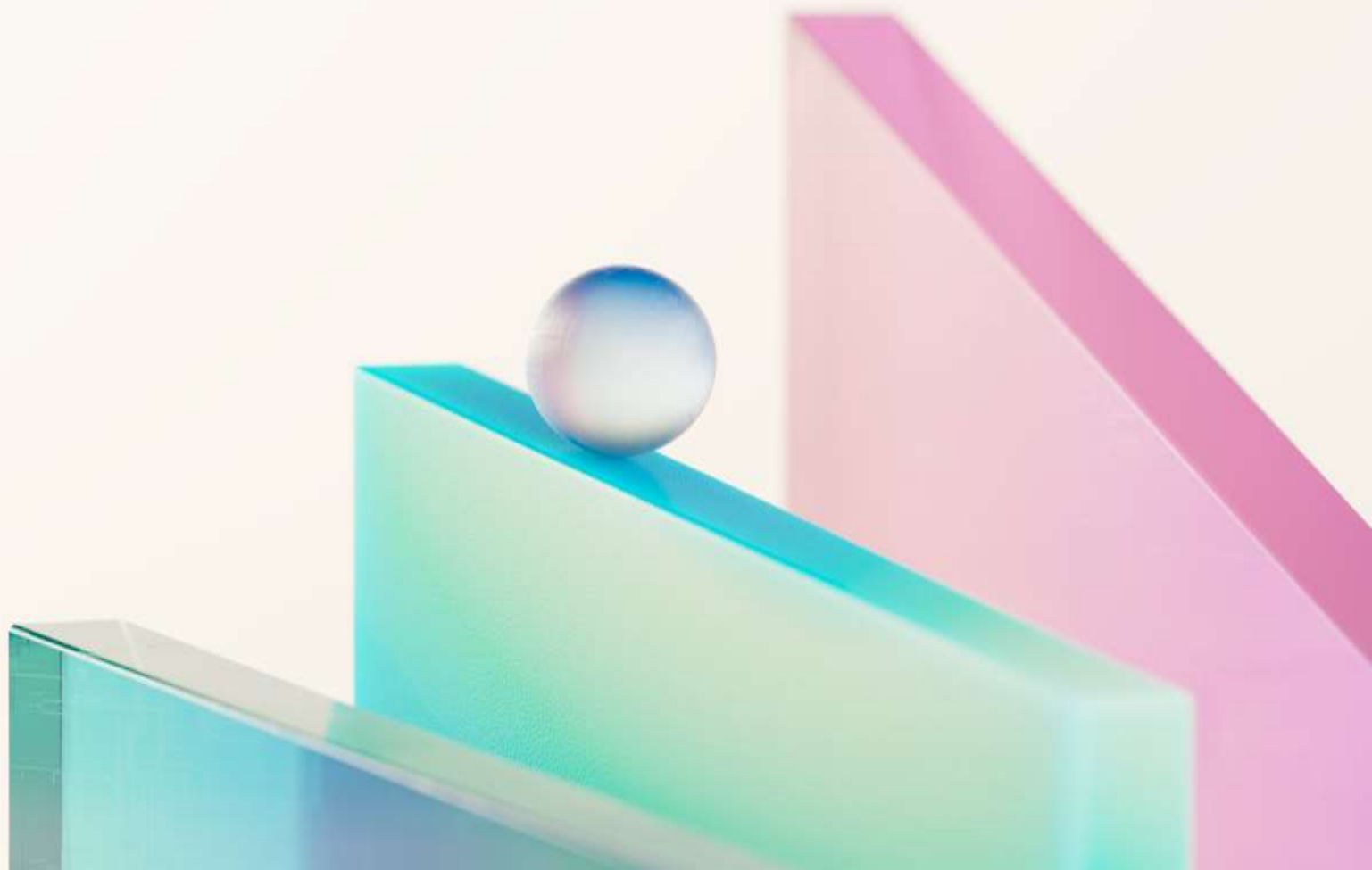
Erwin De Kreuk



Key takeaways

- Automation with PySpark
 - Clean, validate, and deduplicate data with reusable functions
- Business Logic in SQLSpark
 - SQL makes your models portable and easy to maintain.
- Open-Source Productivity
 - Build with our framework to save time and raise quality.
- Code You Can Reuse
 - Try AquaShack yourself via GitHub after the session.

Thank you



Microsoft Fabric
Community Conference

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Engineer Associate – take Exam DP-700
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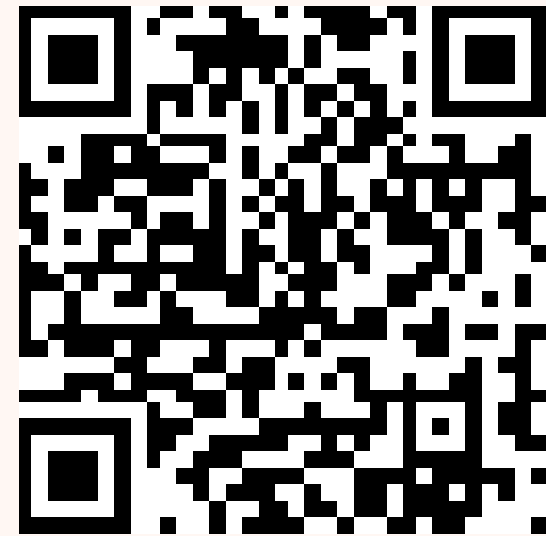
Visit the Fabric Community Lounge to learn
more about this *limited-time offer*.



aka.ms/fabcon/dp700



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Technology experts that share their knowledge and passion with the community